

Name: Ankur De

Roll: BL.SC.U4AIE24015

Files used

Gene.fna

```
>NC_000003.12:46370142-46376206 CCR5 [organism=Homo sapiens]
[GeneID=1234]
[chromosome=3]CTTCAGATAGATTATATCTGGAGTGAAGAATCCTGCCAC
CTATGTATCTGGCATAGTGTGAGTCCTCATAAATGCTTACTGGTTTGAAG
GGCAACAAAATAGTGAACAGAGTGAAAATCCCCACTAAGATCCTGGGTC
CAGAAAAAGATGGGAAACCTGTTTAGCTCACCCGTGAGCCCATAGTTAAA
ACTCTTTAGACAACAGGTTGTTTCCGTTTACAGAGAACAATAATATTGGGT
GGTGAGCATCTGTGTGGGGGTTGGGGTGGGATAGGGGATACGGGGGAGA
GTGGAGAAAAAGGGGACACAGGGTTAATGTGAAGTCCAGGATCCCCCTC
TACATTTAAAGTTG
```

APOE.fna

```
>KY560197.1 Acipenser ruthenus ApoE (apoE) mRNA, partial
cdsCGAGAAGTTGGAACCTACACCGGAAATATCAGAGAGTCCACCCAGC
AGAAGCTGGGCAGCTTGCAGCAGATCTTCTTGGAGAGTGCCGGGAGTCT
CCAAGAGCACTTCAACAGTAAGGCAGAGGAGTTCAAGAAGCA
```

Cftr.fna

```
>KU325498.1 Homo sapiens CFTR (CFTR) gene, partial
cdsTCGTCGTCAATGTGGCACATCGGCTTTTACATGGGCAGCAGAAGGCG
GGGGAAAGAGCAAAAGGAAGGGGTGGTGTGCGGAGTAGGGGTGGGTG
GGGGGAATTGGAAGCAAATGACATCACAGCAGGTCAGAGAAAAAGGGTT
GAGCGGCAGGCACCCAGAGTAGTAGGTCTTTGGCATTAGGAGCTTGAG
CCCAGACGGCCCTAGCAGG
```

Experiment 1

```
def compliment(dna: str):
    dct = {
        'A': 'T',
        'T': 'A',
        'G': 'C',
        'C': 'G' }
    temp = ""
    for c in dna:
        temp = temp + dct[c]

    return temp

with open('gene.fna', 'r', encoding='utf-8') as output_file:
    lines = output_file.readlines()

dna = ""
for line in lines:
    if not line.startswith('>'):
        dna += line

# print(dna)
print(compliment(dna.replace('\n', " ")[:30]))
```

Output:

GAAGTCTATCTAATATAGACCTCACTTCTT

Process finished with exit code 0

Experiment 2

```
from Bio.Seq import Seq

with open('APOE.fna', 'r') as f:
```

```
fl=f.read()

my_dna = Seq(fl)

complement_dna = my_dna.complement()

print(f"Original DNA: {my_dna}")
print(f"Complement DNA: {complement_dna}")
```

Output

Original DNA: >KY560197.1 *Acipenser ruthenus* ApoE (apoE) mRNA, partial cds

CGAGAAGTTGGAACCCTACACCGGAAATATCAGAGAGTCCACCCAGCAGAAGCTGGGCAG
CTTGCAGCAG

ATCTTCTTGGAGAGTGCCGGGAGTCTCCAAGAGCACTTCAACAGTAAGGCAGAGGAGTTC
AAGAAGCA

Complement DNA: >MR560197.1 *Tgipensey yaadenas* TpoE (tpoE) kYNT, pt yaitl ghs

GCTCTTCAACCTTGGGATGTGGCCTTTATAGTCTCTCAGGTGGGTCGTCTTCGACCCGTCG
AACGTCGTC

TAGAAGAACCTCTCACGGCCCTCAGAGGTTCTCGTGAAGTTGTCATTCCGTCTCCTCAAGT
TCTTCGT

Process finished with exit code 0

Experiment 3

```
def compliment(dna: str):
    dct = {
        'A': 'T',
        'T': 'A',
        'G': 'C',
        'C': 'G' }
    temp = ""
    for c in dna:
        temp = temp + dct[c]

    return temp

with open('gene.fna', 'r', encoding='utf-8') as output_file:
    lines = output_file.readlines()

dna = ""
for line in lines:
    if not line.startswith('>'):
        dna += line
dna=dna[:-1]
# print(dna)print(compliment(dna.replace("\n", " ")[:30]))
```

Output

TTATTTTAGAAGGTGTTTTTTATTAGATGC

Process finished with exit code 0

Experiment 4

```
def transcription(dna: str):
    dct = {
        'A': 'T',
        'T': 'A',
        'G': 'C',
        'C': 'G' }
    temp = ""
    for c in dna:
        temp = temp + dct[c]

    return temp

with open('gene.fna', 'r', encoding='utf-8') as output_file:
    lines = output_file.readlines()

dna = ""
for line in lines:
    if not line.startswith('>'):
        dna += line

# print(dna)
print(transcription(dna.replace('\n', ''))[:30])
```

Output:

GUUGTCTUTCTUUTUTUGUCCTCUCTTCTT

Process finished with exit code 0

Experiment 5

```
from Bio.Seq import Seq

with open('cftr.fna', 'r') as f:
    fl=f.read()

my_dna = Seq(fl)
print(my_dna.transcribe())
```

Output

>KU325498.1 Homo sapiens CFUR (CFUR) gene, partial cds

UCGUCGUCAAUGUGGCACAUCGGCUUUUACAUGGGCAGCAGAAGGCGGGGGAAAGAGC
AAAAGGAAGGGG

UGGUGUGCGGAGUAGGGGUGGGUGGGGGGAAUUGGAAGCAAUGACAUCACAGCAGG
UCAGAGAAAAAGG

GUUGAGCGGCAGGCACCCAGAGUAGUAGGUCUUUGGCAUUAGGAGCUUGAGCCCAGAC
GGCCCUAGCAGG

GACCCAGCGCCCGAGAGACCAUGCAGAGGUCGCCUCUGGAAAAGGCCAGCGUUGUCU
CCAAACUUUUUU

UCAGGGGAGAAGGUGGCCAACCGAGCUUCGGAAAGACACGUGCCCACGAAAGAGGAGG
GCGUGUGUAUGG

GUUGGGUUUGGGGUAAAGGAAUAAGCAGUUUUUAAAAAGAUGCGCUAUCAUUCAUUGU
UUUGAAAGAAAA

UGUGGGUAUUGUAGAAUAAAACAGAAAGCAUUAAGAAGAGAUGGAAGAAUGAACUGAAG
CUGAUUGAAUA

GAGAGCCACUCCUACUUGCAAUAGAGAGCCACCAUCCUCCUUGCUCCACAUGUGGCCC
CUUUUCACUUG

UCAUCCUG

Process finished with exit code 0

Experiment 6

```
def to_protein(sequence: str) -> str:
    # Has the one dictionary that stores conversion    rna_ammino = {
        "GCU": "A", "GCC": "A", "GCA": "A", "GCG": "A",
        "UGU": "C", "UGC": "C",
        "GAU": "D", "GAC": "D",
        "GAA": "E", "GAG": "E",
        "UUU": "F", "UUC": "F",
        "GGU": "G", "GGC": "G", "GGA": "G", "GGG": "G",
        "CAU": "H", "CAC": "H",
        "AUU": "I", "AUC": "I", "AUA": "I",
        "AAA": "K", "AAG": "K",
        "UUA": "L", "UUG": "L", "CUU": "L", "CUC": "L", "CUA": "L", "CUG": "L",
        "AUG": "M",
        "AAU": "N", "AAC": "N",
        "CCU": "P", "CCC": "P", "CCA": "P", "CCG": "P",
        "CAA": "Q", "CAG": "Q",
        "CGU": "R", "CGC": "R", "CGA": "R", "CGG": "R", "AGA": "R", "AGG":
    "R",
        "UCU": "S", "UCC": "S", "UCA": "S", "UCG": "S", "AGU": "S", "AGC": "S",
        "ACU": "T", "ACC": "T", "ACA": "T", "ACG": "T",
        "GUU": "V", "GUC": "V", "GUA": "V", "GUG": "V",
        "UGA": "Stop", "UGG": "W",
        "UAU": "Y", "UAC": "Y",
        "UAA": "Stop", "UAG": "Stop",
    }

    ammino=str()

    while sequence:
        try:
            a, b, c, *sequence = sequence
        except Exception as E:
            break
        ammino = ammino + rna_ammino.get(a+b+c, 'X')

    return ammino+'*'
def transcription(dna: str):
```

```

dct = {
    'A': 'T',
    'T': 'U',
    'G': 'C',
    'C': 'G' }
temp = ""
for c in dna:
    temp = temp + dct[c]

return temp

with open('gene.fna', 'r', encoding='utf-8') as output_file:
    lines = output_file.readlines()

dna = ""
for line in lines:
    if not line.startswith('>'):
        dna += line

# print(dna)
print(to_protein(transcription(dna.replace('\n', ""))))

```

Output

```

VXXXXXSXXXWSXXSXSLXGXXSXVLXXXXXXXXXXXXWXXXWSLXXXXXVFXXGLXXXX
XVFXXLLWLXXXXXSLLXXSLPLPLXXPXAXXSXXXXXXLXLWXXGVXXXXSFXXFRXXXXX
CXXXXXFVXVXRXVFXXGLFXXXFRCXXXCFXXXSRFXXXXXXLVXRVXWFXVXXLXXXVS
XPXXXXXXXXXXLXXRAXXXLXXXXCXRXFXVXXXXXXXXLVXLFCGXXXXXXXXVXGGXXXXXF
-----[Not Indicated here]

```

Process finished with exit code 0